

Darshan Singh S

+91-9632880798 | darshans012@gmail.com | darshansingh11.github.io | github.com/Darshansingh11

Education	M.S by Research in Computer Science 2022-2024 <i>International Institute of Information Technology Hyderabad (IIIT-H)</i> CGPA : 9.67/10
Experience	Pre-Doctoral Researcher, Google DeepMind Nov, 2024 - present - Working on Multi-modal and Multi-cultural Video Understanding and Reasoning in Gemini. Machine Learning Engineer, Fast Code AI June, 2024 - Nov, 2024 - Worked with Mercedes-Benz Research & Development to develop a foundation model for the time-series data from Electric Vehicles. - Worked on fine-tuning Stable Diffusion models and adapting them to video games. Research Fellow, CVIT Lab, IIIT Hyderabad July, 2021 - Aug, 2022 - Worked in the Video-Language (Multimodal) team under the supervision of Prof. C. V. Jawahar and Prof. Makarand Tapaswi. - Worked on Audio-Visual lecture segmentation, published at the Winter Conference on Applications of Computer Vision (WACV), 2023.
Technical Skills	Programming Languages - Python, C, C++, MATLAB, Assembly Language(8051), Embedded C Tools/Libraries - PyTorch, NumPy, scikit-learn, $\text{\LaTeX}$
Publications	- Vihari Piratla, Purvam Jain, Darshan Singh S, Trevor Cohn, Partha Talukdar. "Rethinking Cross-lingual Gaps from a Statistical Viewpoint", (under review) 2025. arXiv. - Darshana S[†], Varun Gupta[†], Darshan Singh S[†], Zeeshan Khan, Vineeth Gandhi, Makarand Tapaswi. "VELOCITI: Benchmarking Video-Language Compositional Reasoning with Strict Entailment", In Computer Vision and Pattern Recognition (CVPR) 2025. arXiv. Darshan Singh S, Zeeshan Khan, Makarand Tapaswi. "FiGCLIP: Fine Grained CLIP Adaptation via Densely Annotated Videos", 2024. arXiv. - Manu Gaur, Darshan Singh S, Makarand Tapaswi, "D3: Evaluating MLLM's capacity for Finegrained Visual Discrimination through Self-Retrieval", ECCV-Eval-FoMo Workshop, 2024. view here. - Manu Gaur, Darshan Singh S, Makarand Tapaswi, "No Detail Left Behind: Revisiting Self-Retrieval for Fine-Grained Image Captioning", In Transactions of Machine Learning Research (TMLR) 2024, <i>Top 10%</i>, to be presented at ICLR 2026. arXiv. Darshan Singh S[†], Anchit Gupta[†], C. V. Jawahar, Makarand Tapaswi, "Unsupervised Audio-Visual Lecture Segmentation", Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023. view here
Relevant Courses ^[1]	Statistical Methods in AI*, Topics in Deep Learning*, Advanced NLP, Computer Vision, Data Structures and Algorithms*, Digital Image Processing*, Fairness, Privacy & Ethics in AI*, Operating Systems, Programming in C*, Advanced Calculus*, Multivariable Calculus*, Probability and Statistics,
Selected Projects	FiGCLIP: Fine-Grained CLIP Adaptation via Densely Annotated Videos arXiv - Addressed the lack of fine-grained and syntactic information in CLIP's representations by adapting CLIP on holistic, multidimensional, and densely annotated video-text data. - Proposed a lightweight adaptation strategy with LoRA adapters that enable learning fine-grained representations without catastrophic forgetting. - Performed experiments across 5 diverse tasks: video situation recognition, zero-shot text2video retrieval, zero-shot action recognition, dense video captioning and localization, and VL compositional reasoning and outperformed the base CLIP in all these settings.

^[1] Courses marked (*) indicate grade 'A' awarded for outstanding performance (10/10).

[†] = Equal Contribution

- **VELOCITI: Benchmarking Video-Language Compositional Reasoning with Strict Entailment** [arXiv](#)
 - We propose VELOCITI, a new benchmark to evaluate how well Video-LLMs understand compositional reasoning and also introduce StrictVLE (Strict Video-Language Entailment), a new evaluation method that offers a stricter test by requiring models to correctly classify both positive and negative captions.
 - Our key finding is that even state-of-the-art models like Gemini 1.5 Pro (49.3%) perform far below human accuracy (93.0%), showing they significantly struggle to associate agents with their actions, which is one of the most fundamental reasoning skills.
 - We also provide several key insights, such as understanding actions is harder than agents for open models, Smaller models are predisposed towards ‘Yes’ for the entailment task, etc.
- **No Detail Left Behind - Revisiting Self-Retrieval for Fine-Grained Image Captioning** [arXiv](#)
 - We design- (1) a post-training recipe for self-retrieval finetuning with REINFORCE, and (2) a synthetic framework for visually boosting captioning datasets
 - Jointly, they enable captioners to generate fine-grained, succinct descriptions while reducing hallucinations.
 - Using our training recipe, ClipCap, a 200M param simplification of modern MLLMs, outperforms sota open-source MLLMs on fine-grained visual discrimination.
- **Unsupervised Audio-Visual Lecture Segmentation** [arXiv](#)
 - Proposed video lecture segmentation that splits lectures into bite-sized topics.
 - Approached this problem by first learning the lecture-clip representations by leveraging visual, textual, and OCR cues using a pretext self-supervised task of matching lecture narrations with temporally aligned visual content.
 - Used these learned representations to temporally segment the lectures using an algorithm called TW-FINCH.
 - Introduced a new dataset, AVLectures, a large-scale dataset consisting of 86 courses with over 2,350 lectures covering various STEM subjects from MIT-OpenCourseWare, which we used for pre-training, fine-tuning, and evaluating the segmentation performance.

References

- **IIIT Hyderabad:** [C.V. Jawahar](#), [Makarand Tapaswi](#), [Vineet Gandhi](#)
- **Google DeepMind:** [Partha Talukdar](#), [Shachi Dave](#), [Arsha Nagrani](#)

Letters of recommendation may be provided upon request.